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VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Department of Mathematics

MODULE 4

ORDINARY DIFFERENTIAL EQUATIONS OF HIGHER ORDER

Linear ODEs with Constant Coefficients	Inverse Differential Operator (PI)	Cauchy's & Legendre's Equations
Method of Variation of Parameters	L-C & L-C-R Circuit Applications	Previous Year VTU Questions

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1. FORMULAE & STANDARD RESULTS QUICK REFERENCE

IMPORTANT: Memorise every formula in this section — they appear **DIRECTLY** in VTU exam questions!

1.1 Summary: CF Rules by Root Type

Root Type	Roots of A.E.	Complementary Function (CF)
Real & Distinct	$m_1, m_2, \dots, m_n$	$y = C_1 e^{(m_1 x)} + C_2 e^{(m_2 x)} + \dots + C_n e^{(m_n x)}$
Real & Equal ($m_1=m_2=m_3=m$)	m, m, m (triple)	$y = (C_1 + C_2 x + C_3 x^2) e^{(m x)}$
Complex: $\alpha \pm i\beta$	$\alpha + i\beta, \alpha - i\beta$	$y = e^{(\alpha x)} [C_1 \cos(\beta x) + C_2 \sin(\beta x)]$
Two pairs complex (repeated)	$\alpha \pm i\beta$ (twice)	$y = e^{(\alpha x)} [(C_1 + C_2 x) \cos(\beta x) + (C_3 + C_4 x) \sin(\beta x)]$
Pure imaginary	$\pm i\beta$	$y = C_1 \cos(\beta x) + C_2 \sin(\beta x)$

1.2 Summary: PI Rules by $\phi(x)$ Type

$\phi(x)$ Form	When $f(a) \neq 0$	When $f(a) = 0$ (failure case)
$e^{(ax)}$	$PI = e^{(ax)} / f(a)$ [put $D=a$]	$PI = x e^{(ax)} / f'(a)$ [if $f(a) \neq 0$; $x^2 e^{(ax)} / f''(a)$...]
$\sin(ax)$ or $\cos(ax)$	$PI = \sin(ax) / f(-a^2)$ [put $D^2 = -a^2$]	$PI = x \sin(ax) / f'(-a^2)$ [failure when $f(-a^2) = 0$]
x^n (polynomial)	$PI = [f(D)]^{-1} x^n$ [expand by long division]	No failure — always works
$e^{(ax)} * V(x)$	$PI = e^{(ax)} * [1/f(D+a)] * V(x)$	Shift the operator, then solve for $V(x)$
k (constant)	$PI = k/f(0)$ [put $D=0$]	$PI = k x / f'(0)$ if $f(0) = 0$

1.3 Key Standard Integrals and Identities

#	Formula / Identity	Use in ODE Problems
1	$\int e^{(ax)} \cos(bx) dx = e^{(ax)} / (a^2 + b^2) [a \cos(bx) + b \sin(bx)]$	Used in PI for $e^{(ax)} * \text{trig}$
2	$\int e^{(ax)} \sin(bx) dx = e^{(ax)} / (a^2 + b^2) [a \sin(bx) - b \cos(bx)]$	Used in PI for $e^{(ax)} * \text{trig}$
3	$\cosh(ax) = (e^{(ax)} + e^{(-ax)}) / 2$	Convert to exponentials for PI
4	$\sinh(ax) = (e^{(ax)} - e^{(-ax)}) / 2$	Convert to exponentials for PI
5	$\cos^2(x) = (1 + \cos 2x) / 2$	Polynomial-trig product type
6	$\sin^2(x) = (1 - \cos 2x) / 2$	Polynomial-trig product type

7	$4\cos^3(x) = \cos(3x) + 3\cos(x)$	Polynomial-trig product type
8	$1/(D)$ means integration w.r.t. x	Inverse operator as integrator

2. LINEAR ODEs OF HIGHER ORDER — FUNDAMENTALS

2.1 Definition and Standard Form

A Differential Equation (DE) in which the dependent variable y and all its derivatives appear only in the first degree (not raised to any power) and there are no products of y with its derivatives is called a Linear DE.

A linear DE of the n -th order with constant coefficients has the general form:

$$\left| \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + a_2 \frac{d^{n-2} y}{dx^{n-2}} + \dots + a_{n-1} \frac{dy}{dx} + a_n y = \phi(x) \right|$$

where $a_1, a_2, \dots, a_n$ are ALL CONSTANTS and $\phi(x)$ is a function of x alone (or a constant).

NOTE: If $\phi(x) = 0$, the equation is called a Linear HOMOGENEOUS DE and its solution is called the Complementary Function (CF). If $\phi(x) \neq 0$, it is NON-HOMOGENEOUS and the complete solution = CF + Particular Integral (PI).

2.2 Operator Notation D

To simplify writing, we use the differential operator D where:

OPERATOR D

$$D = d/dx \quad D^2 = d^2/dx^2 \quad D^n = d^n/dx^n$$

Using D -notation, the general n -th order linear ODE becomes:

$$\left| f(D) * y = \phi(x) \quad \text{where } f(D) = D^n + a_1 D^{n-1} + \dots + a_{n-1} D + a_n \right|$$

For the homogeneous case $\phi(x) = 0$: $f(D) * y = 0$

NOTE: The key advantage: we can treat $f(D)$ algebraically, factoring it, finding roots, and using algebraic shortcuts to find PI. This is the power of the operator method.

2.3 Complementary Function (CF) — Finding Method

Steps to find the Complementary Function:

S1

Write the Auxiliary Equation (A.E.)

Replace D by m in $f(D)$ to get $f(m) = 0$. This gives the Auxiliary Equation (A.E.) or Characteristic Equation

S2

Solve the A.E. for roots $m_1, m_2, \dots, m_n$

Use factoring, synthetic division (Horner's method), or quadratic formula. If degree ≥ 3 : try $m = \pm 1, \pm 2, \pm 1/2$ by inspection first

S3

Write CF based on the nature of roots

Apply the appropriate case (I, II, or III) given below

S4**Add all parts**If there are n roots, CF contains n arbitrary constants $C_1, C_2, \dots, C_n$ **2.4 Case I — Real and Distinct Roots**If the roots of the A.E. are $m_1, m_2, m_3, \dots, m_n$ (all real and different), then:**CASE I CF**

$$CF = C_1 e^{(m_1 x)} + C_2 e^{(m_2 x)} + C_3 e^{(m_3 x)} + \dots + C_n e^{(m_n x)}$$

NOTE: Each distinct root m_i gives one term $C_i e^{(m_i x)}$. Total number of arbitrary constants = order of the ODE.

2.5 Case II — Real and Equal RootsIf two or more roots are equal, say $m_1 = m_2 = m$ (repeated root of multiplicity 2), then:**CASE II (double root)**

$$CF = (C_1 + C_2 x) e^{(m x)} \quad [for\ m\ repeated\ twice]$$

If $m_1 = m_2 = m_3 = m$ (triple root):**CASE II (triple root)**

$$CF = (C_1 + C_2 x + C_3 x^2) e^{(m x)} \quad [for\ m\ repeated\ thrice]$$

NOTE: Pattern: If a root m is repeated k times, the corresponding part of CF is $(C_1 + C_2 x + C_3 x^2 + \dots + C_k x^{(k-1)}) e^{(m x)}$

2.6 Case III — Complex Conjugate RootsIf two roots are complex: $m = \alpha \pm i\beta$ (they always occur in conjugate pairs for real-coefficient ODEs), then:**CASE III (complex roots)**

$$CF = e^{(\alpha x)} [C_1 \cos(\beta x) + C_2 \sin(\beta x)]$$

If there are two DISTINCT pairs of complex roots, say $\alpha \pm i\beta$ and $\gamma \pm i\delta$:

$$CF = e^{(\alpha x)} [C_1 \cos(\beta x) + C_2 \sin(\beta x)] + e^{(\gamma x)} [C_3 \cos(\delta x) + C_4 \sin(\delta x)]$$

If $\alpha \pm i\beta$ is a REPEATED pair (occurring twice):**CASE III (repeated complex)**

$$CF = e^{(\alpha x)} [(C_1 + C_2 x) \cos(\beta x) + (C_3 + C_4 x) \sin(\beta x)]$$

NOTE: Special case: Pure imaginary roots $\pm i\beta$ ($\alpha=0$). Then $CF = C_1 \cos(\beta x) + C_2 \sin(\beta x)$ — no exponential factor!

2.7 Worked Examples — CF Problems

Example 1: Solve $d^3y/dx^3 - 3d^2y/dx^2 + 3dy/dx - y = 0$

Given: $(D^3 - 3D^2 + 3D - 1)y = 0$

A.E.

Write and solve: $m^3 - 3m^2 + 3m - 1 = 0$

By inspection, $m = 1$ is a root. Synthetic division: $m=1 \mid 1 \ -3 \ 3 \ -1 \mid 0 \ 1 \ -2 \ 1 \mid 1$
 $-2 \ 1 \ 0$ Reduced: $m^2 - 2m + 1 = (m-1)^2 = 0 \Rightarrow m = 1, 1$ All 3 roots: $m = 1, 1, 1$ (triple repeated root)

SOLUTION

$$CF = y = (C_1 + C_2x + C_3x^2)e^x$$

Example 2: Solve $(4D^4 - 4D^3 - 23D^2 + 12D + 36)y = 0$

Given 4th-order ODE. A.E.: $4m^4 - 4m^3 - 23m^2 + 12m + 36 = 0$

A.E.

Factor the 4th-degree polynomial

By inspection $m=2$: Synthetic division (dividing $4 \ -4 \ -23 \ 12 \ 36$ by $(m-2)$): $m=2 \mid 4 \ -4 \ -23 \ 12 \ 36 \mid 8 \ 8 \ -30 \ -36 \mid 4 \ 4 \ -15 \ -18 \ 0$ Quotient: $4m^3 + 4m^2 - 15m - 18 = 0$ Again $m=2$ by inspection: $4 \ 4 \ -15 \ -18$ by $(m-2)$: $\mid 0 \ 8 \ 24 \ 18 \mid 4 \ 12 \ 9 \ 0$ Quotient: $4m^2 + 12m + 9 = (2m+3)^2 = 0 \Rightarrow m = -3/2$ (double)

Four roots: $m = 2, 2, -3/2, -3/2$ (two pairs of repeated roots)

SOLUTION

$$y = (C_1 + C_2x)e^{2x} + (C_3 + C_4x)e^{-3x/2}$$

Example 3: Solve $(4D^4 - 8D^3 - 7D^2 + 11D + 6)y = 0$

A.E.: $4m^4 - 8m^3 - 7m^2 + 11m + 6 = 0$

A.E.

Step-by-step synthetic division

Try $m=-1$: $4 \ -8 \ -7 \ 11 \ 6 \mid 0 \ -4 \ 12 \ -5 \ -6 \mid 4 \ -12 \ 5 \ 6 \ 0$ $4m^3 - 12m^2 + 5m + 6 = 0$ Try $m=2$: $4 \ -12 \ 5 \ 6 \mid 0 \ 8 \ -8 \ -6 \mid 4 \ -4 \ -3 \ 0$ $4m^2 - 4m - 3 = 0 \Rightarrow (2m+1)(2m-3)=0 \Rightarrow m=-1/2, m=3/2$ Four distinct roots: $-1, 2, -1/2, 3/2$

SOLUTION

$$CF = C_1e^{-x} + C_2e^{2x} + C_3e^{-x/2} + C_4e^{3x/2}$$

Example 4: Solve $(D^4 - m^4)y = 0$ (note: m is parameter, not root variable)

A.E.

A.E. is $p^4 - m^4 = 0$ (using p for the dummy variable to avoid confusion)

$(p^2+m^2)(p^2-m^2) = 0$ $p^2+m^2 = 0 \Rightarrow p = \pm m \cdot i$ (complex, $\alpha=0, \beta=m$) $p^2-m^2 = 0 \Rightarrow p = \pm m$ (real distinct)

SOLUTION

$$y = C_1e^{mx} + C_2e^{-mx} + e^{0}[C_3\cos(mx) + C_4\sin(mx)] \Rightarrow y = C_1e^{mx} + C_2e^{-mx} + C_3\cos(mx) + C_4\sin(mx)$$

Example 5: Solve $(D^4 + 64)y = 0$

A.E.	$m^4 + 64 = 0$. Use: $a^2 + b^2 = (a+b)^2 - 2ab$ with $a=m^2$, $b=8$ $(m^2+8)^2 - 16m^2 = 0$ $(m^2+8+4m)(m^2+8-4m) = 0$ $m^2+4m+8=0$: $m = \frac{-4 \pm \sqrt{16-32}}{2} = \frac{-4 \pm 4i}{2} = -2 \pm 2i$ $m^2-4m+8=0$: $m = \frac{4 \pm \sqrt{16-32}}{2} = \frac{4 \pm 4i}{2} = 2 \pm 2i$
SOLUTION	$y = e^{-2x} [C1 \cos(2x) + C2 \sin(2x)] + e^{2x} [C3 \cos(2x) + C4 \sin(2x)]$

Example 6: IVP — Solve $d^2x/dt^2 + 5dx/dt + 6x = 0$, $x(0)=0$, $x'(0)=15$

S1	A.E.: $m^2 + 5m + 6 = 0$ $m^2 + 3m + 2m + 6 = 0 \Rightarrow (m+3)(m+2) = 0 \Rightarrow m = -2, -3$ CF = $C1e^{-2t} + C2e^{-3t}$... (general solution)
S2	Differentiate the general solution $dx/dt = -2C1e^{-2t} - 3C2e^{-3t}$
S3	Apply IC1: $x=0$ at $t=0$ $0 = C1e^0 + C2e^0 = C1 + C2 \Rightarrow C1 + C2 = 0$... (3)
S4	Apply IC2: $dx/dt=15$ at $t=0$ $15 = -2C1 - 3C2$... (4) From (3): $C1 = -C2$. Sub in (4): $15 = -2(-C2) - 3C2 = 2C2 - 3C2 = -C2 \Rightarrow C2 = -15$, $C1 = 15$
SOLUTION	$x(t) = 15e^{-2t} - 15e^{-3t}$

3. PARTICULAR INTEGRAL (PI) — INVERSE OPERATOR METHOD

For the non-homogeneous ODE $f(D)y = \phi(x)$, the complete General Solution (GS) is:

GENERAL SOLUTION	$GS = y = CF + PI \quad \text{where } PI = [1/f(D)] * \phi(x)$
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The inverse operator $[1/f(D)]$ is used to find the PI. The method depends entirely on the form of $\phi(x)$. There are 4 standard types used in VTU syllabus.

3.1 General Method for PI

The Particular Integral is:

$$PI = \phi(x) / f(D) \quad [\text{inverse differential operator applied to } \phi(x)]$$

NOTE: The key principle: $1/f(D)$ means we need to find a function y_p such that $f(D)y_p = \phi(x)$. We do NOT literally divide — we use algebraic rules based on the type of $\phi(x)$.

3.2 Type 1: $\phi(x) = e^{ax}$ — Exponential RHS

This is the most common type. Put $D = a$ in $f(D)$:

TYPE 1 RULE	$PI = (1/f(D)) * e^{ax} = e^{ax}/f(a) \quad [\text{provided } f(a) \neq 0]$
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FAILURE CASES — when $f(a) = 0$:

NOTE: If $f(a) = 0$ [denominator = 0], multiply by x and differentiate the denominator: $PI = x * e^{ax}/f'(a)$. If $f'(a) = 0$ again, $PI = x^2 * e^{ax}/f''(a)$, and so on.

IMPORTANT: Exponential rule also applies for:

- $\phi(x) = a^x = e^{x \log a} \rightarrow$ put $a = \log(a)$ in the formula
- $\phi(x) = \cosh(ax) = (e^{ax} + e^{-ax})/2 \rightarrow$ split and apply to each exponential
- $\phi(x) = \sinh(ax) = (e^{ax} - e^{-ax})/2 \rightarrow$ split and apply to each exponential

Example 1: Solve $(D^2 - 6D + 9)y = 6e^{3x}$

CF	A.E.: $m^2 - 6m + 9 = 0 \Rightarrow (m-3)^2 = 0 \Rightarrow m = 3, 3$ (equal) CF = $(C_1 + C_2 x) * e^{3x}$
PI	PI = $6e^{3x}/(D^2 - 6D + 9)$. Put $D=3$: $f(3) = 9 - 18 + 9 = 0$ [FAILURE! denominator = 0] $\Rightarrow PI = 6 * x * e^{3x}/f'(D) _{D=3}$ where $f'(D) = 2D - 6$ $f'(3) = 6 - 6 = 0$ [FAILURE again!] $\Rightarrow PI = 6 * x^2 * e^{3x}/f''(D) _{D=3}$ where $f''(D) = 2$ PI = $6 * x^2 * e^{3x}/2 = 3x^2 * e^{3x}$

SOLUTION	$y = (C_1 + C_2 x) e^{(3x)} + 3x^2 e^{(3x)}$
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Example 2: Solve $(D^2 - 4D + 13)y = e^{(3x)} \cosh(2x) + 3^x$

CF	A.E.: $m^2 - 4m + 13 = 0$. $m = (4 \pm \sqrt{16 - 52})/2 = (4 \pm 6i)/2 = 2 \pm 3i$ CF = $e^{(2x)} [C_1 \cos(3x) + C_2 \sin(3x)]$
PI-P 1	For $e^{(3x)} \cosh(2x)$: convert $\cosh(2x) = (e^{(2x)} + e^{(-2x)})/2$ PI1 = $[1/(D^2 - 4D + 13)] * e^{(3x)} * \{(e^{(2x)} + e^{(-2x)})/2\} = (1/2) * [e^{(5x)}/(25 - 20 + 13) + e^{(x)}/(1 - 4 + 13)] = (1/2) * [e^{(5x)}/18 + e^{(x)}/10]$
PI-P 2	For $3^x = e^{(x \log 3)}$. Put $D = \log 3$: PI2 = $3^x / [(\log 3)^2 - 4(\log 3) + 13]$
SOLUTION	$y = e^{(2x)} [C_1 \cos 3x + C_2 \sin 3x] + e^{(5x)}/36 + e^{(x)}/20 + 3^x / [(\log 3)^2 - 4(\log 3) + 13]$

3.3 Type 2: $\phi(x) = \sin(ax)$ or $\cos(ax)$ — Trigonometric RHS

For $\sin(ax)$ or $\cos(ax)$, replace D^2 with $-a^2$ in $f(D^2)$:

TYPE 2 RULE	$PI = \sin(ax) / f(D^2) = \sin(ax) / f(-a^2) \quad [\text{provided } f(-a^2) \neq 0]$
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NOTE: Same rule for $\cos(ax)$: $PI = \cos(ax) / f(-a^2)$. If $f(D)$ contains both D and D^2 , separate them. The rule $D^2 \rightarrow -a^2$ applies only to the D^2 terms; use conjugate multiplication for remaining D terms.

Also applies to $\sin(ax+b)$ and $\cos(ax+b)$ — same rule with $D^2 \rightarrow -a^2$.

FAILURE CASE: If $f(-a^2) = 0$:

NOTE: When $f(-a^2) = 0$, $PI = x * \sin(ax) / f'(-a^2)$ [for cosine multiply by x and integrate]

Example 1: Solve $y'' - 4y' + 4y = 8\cos(2x)$

CF	A.E.: $m^2 - 4m + 4 = 0 \Rightarrow (m-2)^2 = 0 \Rightarrow m=2, 2$ CF = $(C_1 + C_2 x) e^{(2x)}$
PI	$PI = 8\cos(2x) / (D^2 - 4D + 4)$. $D^2 \rightarrow -2^2 = -4$: $f(-4) = -4 - 4D + 4 = -4D \Rightarrow PI = 8\cos(2x) / (-4D) = -2 * (1/D) * \cos(2x) = -2 * \int \cos(2x) dx = -2 * \sin(2x) / 2 = -\sin(2x)$
SOLUTION	$y = (C_1 + C_2 x) e^{(2x)} - \sin(2x)$

Example 2: Solve $(D^2 - 3D + 2)y = \sin(2x)$

CF	A.E.: $m^2-3m+2=0 \Rightarrow (m-1)(m-2)=0 \Rightarrow m=1,2$ CF = $C_1 e^x + C_2 e^{2x}$
PI	$D^2 \rightarrow -4$. $f(-4) = -4-3D+2 = -2-3D$ PI = $\sin(2x)/(-2-3D)$. Rationalize by multiplying top/bottom by $(-2+3D)$: $= (-2+3D)\sin(2x) / [(-2)^2-(3D)^2] = (-2+3D)\sin(2x) / (4-9D^2)$ $D^2 \rightarrow -4$ again: $4-9(-4) = 4+36 = 40$ PI = $[(-2)\sin(2x) + 3D(\sin(2x))] / 40$ $D(\sin(2x)) = 2\cos(2x)$ PI = $[-2\sin(2x) + 6\cos(2x)] / 40 = [3\cos(2x) - \sin(2x)]/20$ Use $1/(2D)$: PI = $(1/2D)[3\cos(2x) - \sin(2x)] \rightarrow$ integrate: $= (1/2)[3\sin(2x)/2 - (-\cos(2x)/2)] * \dots$ let me finalize via earlier working PI = $(1/2D)\{3\cos(2x) - \sin(2x)\} = [3\sin(2x)/2 + \cos(2x)/2] / \dots$ = final boxed above

Correct final: using the rationalized denominator 40: PI = $(3\cos(2x) - \sin(2x))/20$, then $1/D$:

SOLUTION	$y = C_1 e^x + C_2 e^{2x} + (1/20D) [3\cos(2x) - \sin(2x)]$
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Example 3: Solve $(D^2-4D+4)y = e^{2x} + \sin(x)$

CF	A.E.: $(m-2)^2=0 \Rightarrow m=2,2 \Rightarrow CF=(C_1+C_2 x)e^{2x}$
PI1	For e^{2x}: $D \rightarrow 2$, denom=0. Use failure rule twice. PI1 = $x^2 e^{2x}/2$
PI2	For $\sin(x)$: $D^2 \rightarrow -1$, $f(-1)=-1-4D+4=3-4D$ PI2 = $\sin(x)/(3-4D)$. Rationalize with $(3+4D)$: $= (3+4D)\sin(x)/(9-16D^2)$. $D^2 \rightarrow -1$: $9+16=25$. $= [3\sin(x)+4\cos(x)]/25$
SOLUTION	$y = (C_1+C_2 x) e^{2x} + x^2 e^{2x}/2 + (3\sin x + 4\cos x)/25$

3.4 Type 3: $\phi(x) = x^n$ (Polynomial) — Long Division Method

When $\phi(x)$ is a polynomial in x , we expand $[1/f(D)]$ as a power series in D using the binomial expansion, then apply to x^n .

TYPE 3 RULE	$PI = [1/f(D)] * x^n$. Write $f(D) = a_0[1+g(D)]$, so $PI = (1/a_0) * [1+g(D)]^{-1} * x^n$
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NOTE: The binomial expansions most commonly used: $(1+u)^{-1} = 1 - u + u^2 - u^3 + \dots$ $(1-u)^{-1} = 1 + u + u^2 + u^3 + \dots$ STOP when the degree exceeds n (all higher derivatives of x^n are zero)

ALTERNATIVE: Perform polynomial long division (as shown in the handwritten notes). Divide x^n by $f(D)$ treating D as the divisor variable.

Example 1: Solve $(D^2+4)y = x^2$

CF	$m^2+4=0 \Rightarrow m=\pm 2i \Rightarrow CF = C_1 \cos(2x) + C_2 \sin(2x)$
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PI	$PI = x^2/(D^2+4) = (1/4)*x^2/(1+D^2/4)$ $= (1/4)*(1+D^2/4)^{-1}*x^2 = (1/4)*(1 - D^2/4 + \dots)*x^2 = (1/4)*(x^2 - 2/4) = x^2/4 - 1/8$
SOLUTION	$y = C1*\cos(2x) + C2*\sin(2x) + x^2/4 - 1/8$

Example 2: Solve $y'' + 2y' + y = 2x + x^2$

CF	$(D^2+2D+1)y=(D+1)^2*y=0 \Rightarrow m=-1,-1 \Rightarrow CF=(C1+C2*x)*e^{-x}$
PI	$PI = (x^2+2x)/(D^2+2D+1) = (x^2+2x)/(1+2D+D^2)$ Long division: divide x^2+2x by $1+2D+D^2$ (treating D as variable): x^2+2x Quotient $= x^2-2x+2$ $-(x^2+4x+2) \rightarrow -(x^2+4x+2) = -2x-2$ $-(-2x-4) = -(-2x+(-4)) \rightarrow -(-2x-4+2) = 2 \rightarrow$ etc. $PI = x^2 - 2x + 2$
SOLUTION	$y = (C1+C2*x)*e^{-x} + x^2 - 2x + 2$

3.5 Type 4: PI when $\phi(x) = e^{ax}*V(x)$ — Shift Theorem

When the RHS is a product of an exponential and another function $V(x)$:

SHIFT THEOREM	$(1/f(D)) * e^{ax} * V(x) = e^{ax} * [1/f(D+a)] * V(x)$
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NOTE: The exponential 'shifts' the operator from $f(D)$ to $f(D+a)$. Then find $[1/f(D+a)]*V(x)$ using Type 1, 2, or 3 as appropriate for $V(x)$.

Example: Solve $(D^2-2D+4)y = e^x*\cos(x)$

CF	$A.E.: m^2-2m+4=0. m=(2\pm\sqrt{4-16})/2 = 1\pm i*\sqrt{3}$ $CF = e^x[C1*\cos(\sqrt{3}x) + C2*\sin(\sqrt{3}x)]$
PI	$\text{Shift theorem: } a=1, V(x)=\cos(x). f(D+1) = (D+1)^2-2(D+1)+4$ $= D^2+2D+1-2D-2+4 = D^2+3$ $PI = e^x * \cos(x)/(D^2+3). D^2 \rightarrow -1: f(-1)=-1+3=2$ $PI = e^x*\cos(x)/2$
SOLUTION	$y = e^x[C1*\cos(\sqrt{3}x) + C2*\sin(\sqrt{3}x)] + (e^x*\cos(x))/2$

3.6 Comprehensive Worked Examples — Various Types

Example 1: Solve $(D^3+6D^2+11D+6)y = e^x + 1$

CF	$m^3+6m^2+11m+6=0. \text{ Roots: } m=-1,-2,-3 \text{ (by inspection and Horner's)}$ $CF = C1*e^{-x} + C2*e^{-2x} + C3*e^{-3x}$
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PI	$PI = (e^x + 1)/(D^3 + 6D^2 + 11D + 6) = P_1 + P_2$
P1	For e^x : $D \rightarrow 1$. $f(1) = 1 + 6 + 11 + 6 = 24$ $P_1 = e^x/24$
P2	For $1 = e^{(0 \cdot x)}$: $D \rightarrow 0$. $f(0) = 0 + 0 + 0 + 6 = 6$ $P_2 = 1/6$
SOLUTION	$y = C_1 * e^{-x} + C_2 * e^{-2x} + C_3 * e^{-3x} + e^x/24 + 1/6$

Example 2: Solve $(D^3 - D^2 + 4D - 4)y = \sinh(2x + 3)$

CF	$m^3 - m^2 + 4m - 4 = 0$. $m = 1 \pm 2i$ and $m = 1$ (by grouping or Horner's) $CF = C_1 * e^x + e^0[C_2 * \cos(2x) + C_3 * \sin(2x)] = C_1 * e^x + C_2 * \cos(2x) + C_3 * \sin(2x)$
PI	$\sinh(2x + 3) = (e^{(2x+3)} - e^{-(2x+3)})/2 = (e^3 * e^{2x} - e^{-3} * e^{-2x})/2$ $P_1 = (e^3/2) * e^{2x} / (D^3 - D^2 + 4D - 4)$. $D \rightarrow 2$: $f(2) = 8 - 4 + 8 - 4 = 8$ $P_1 = e^3 * e^{2x} / (2 * 8) = e^{(2x+3)}/16$ $P_2 = (e^{-3}/2) * e^{-2x}$. $D \rightarrow -2$: $f(-2) = -8 - 4 - 8 - 4 = -24$ $P_2 = e^{-3} * e^{-2x} / (2 * (-24)) = -e^{-(2x+3)}/48$ $PI = e^{(2x+3)}/16 + e^{-(2x+3)}/48$
SOLUTION	$y = C_1 * e^x + C_2 * \cos 2x + C_3 * \sin 2x + e^{(2x+3)}/16 + e^{-(2x+3)}/48$

Example 3: Solve $(D^2 + 3D + 2)y = 4\cos^2(x)$

Convert	$\cos^2(x) = (1 + \cos 2x)/2$. So $RHS = 4 * (1 + \cos 2x)/2 = 2(1 + \cos 2x) = 2e^{(0x)} + 2\cos 2x$
CF	$m^2 + 3m + 2 = 0 \Rightarrow (m+1)(m+2) = 0 \Rightarrow m = -1, -2 \Rightarrow CF = C_1 * e^{-x} + C_2 * e^{-2x}$
PI	$P_1 = 2 * e^{(0x)} / (D^2 + 3D + 2)$. $D \rightarrow 0$: $f(0) = 2$. $P_1 = 2/2 = 1$ $P_2 = 2\cos(2x) / (D^2 + 3D + 2)$. $D^2 \rightarrow -4$: $(-4 + 3D + 2) = (3D - 2) = 2(3D + 2)\cos(2x) / ((3D - 2)(3D + 2)) = 2(3D + 2)\cos(2x) / (9D^2 - 4)$ $D^2 \rightarrow -4$: $9(-4) - 4 = -40 = 2(3D + 2)\cos(2x) / (-40) = -(6D(\cos 2x) + 4\cos 2x) / 40 = -(12\sin 2x + 4\cos 2x) / 40 = (12\sin 2x - 4\cos 2x) / 40 = (3\sin 2x - \cos 2x) / 10$
SOLUTION	$y = C_1 * e^{-x} + C_2 * e^{-2x} + 1 + (3\sin 2x - \cos 2x) / 10$

Example 4: Solve $(D - 2)^2 y = 8(e^{2x} + \sin(2x) + x^2)$

CF	$(m - 2)^2 = 0 \Rightarrow m = 2, 2 \Rightarrow CF = (C_1 + C_2 * x) * e^{2x}$
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P1 = P1+ P2+ P3	$P1 = 8e^{(2x)}/(D^2-4D+4)$. $D \rightarrow 2$: $f(2)=0$. Failure twice. $P1 = 8x^2e^{(2x)}/f'(2) = 8x^2e^{(2x)}/2 = 4x^2e^{(2x)}$
P2	For $8\sin(2x)$: $D^2 \rightarrow -4$. $f(-4) = -4-4D+4 = -4D$ $P2 = 8\sin(2x)/(-4D) = -2(1/D)\sin(2x) = -2(-\cos 2x/2) = \cos(2x)$
P3	For $8x^2/(D^2-4D+4)$: divide $8x^2$ by $(4-4D+D^2)$ by long division $8x^2/(4-4D+D^2)$: Leading term $8x^2/4=2x^2$ $4*2x^2=8x^2$; $-4D(2x^2)=-16x$; $D^2(2x^2)=4$ Remainder: $8x^2-(8x^2-16x+4)=16x-4$ Next: $(16x-4)/4=4x$; $4*4x=16x$; $-4D(4x)=-16$ Remainder: $16x-4-(16x-16)=12$ Next: $12/4=3$ So $P3=2x^2+4x+3$
SOL UTIO N	$y = (C1+C2*x)*e^{(2x)} + 4x^2*e^{(2x)} + \cos(2x) + 2x^2+4x+3$

4. LINEAR ODEs WITH VARIABLE COEFFICIENTS

When the coefficients of y and its derivatives are functions of x (not constants), the operator method fails. Two special types are reducible to constant-coefficient ODEs via substitution.

4.1 Cauchy's Linear Differential Equation

An ODE of the form:

CAUCHY'S EQUATION

$$a_0 x^n y^{(n)} + a_1 x^{(n-1)} y^{(n-1)} + \dots + a_{(n-1)} x y' + a_n y = \phi(x)$$

where $a_0, a_1, \dots, a_n$ are CONSTANTS (not functions of x), is called Cauchy's Linear DE (or Euler-Cauchy Equation).

SOLVING METHOD — Substitution:

SUBSTITUTION

Put $t = \log(x)$ OR equivalently $e^t = x$ (so $x = e^t$)

This transforms $x^k y^{(k)}$ into expressions with D-operator w.r.t. t

TRANSFORMATION RULES:

$$x y' = D y$$

$$(where D = d/dt)$$

$$x^2 y'' = D(D-1)y$$

$$x^3 y''' = D(D-1)(D-2)y$$

NOTE: Pattern: $x^n y^{(n)} = D(D-1)(D-2)\dots(D-n+1)y$. After substitution, the ODE becomes a constant-coefficient ODE in t . Solve for y in terms of t , then back-substitute $t = \log(x)$.

4.2 Legendre's Linear Differential Equation

An ODE of the form:

LEGENDRE'S EQUATION

$$a_0 (ax+b)^n y^{(n)} + a_1 (ax+b)^{(n-1)} y^{(n-1)} + \dots + a_n y = \phi(x)$$

SOLVING METHOD — Substitution:

SUBSTITUTION

Put $t = \log(ax+b)$ OR $e^t = ax+b \Rightarrow x = (e^t - b)/a$

Transformation rules (where a is coefficient of x in $(ax+b)$):

$$(ax+b)y' = aDy$$

$$D = d/dt$$

$$(ax+b)^2 y'' = a^2 D(D-1)y$$

$$(ax+b)^3 y''' = a^3 D(D-1)(D-2)y$$

NOTE: Legendre's equation is the GENERAL CASE. Cauchy's equation is a special case with $a=1$, $b=0$ (so $(ax+b) = x$). Use the same transformation technique with $e^t = ax+b$.

4.3 Worked Examples — Cauchy's Equation

Example 1: Solve $x^2 y'' - x y' + y = \log(x)$

S1	Identify as Cauchy's DE. Put $t=\log(x)$, $e^t=x$ $xy' = Dy$, $x^2 y'' = D(D-1)y$, where $D = d/dt$
S2	Substitute into ODE $D(D-1)y - Dy + y = t$ $(D^2-D-D+1)y = t$ $(D^2-2D+1)y = t$... (ODE in t with constant coeff)
CF	A.E.: $m^2-2m+1=0 \Rightarrow (m-1)^2=0 \Rightarrow m=1,1$ CF = $(C_1+C_2 t)e^t$
PI	PI = $t/(1-2D+D^2)$. Divide t by $(1-2D+D^2)$ by long division: Quotient = $t+2$ (verify: $(1-2D+D^2)(t+2) = t+2 - 2*1 + 0 = t$ ✓) PI = $t+2$
S3	$y = CF+PI = (C_1+C_2 t)e^t + t+2$. Back-sub $t=\log(x)$, $e^t=x$: $y = (C_1+C_2 \log(x))x + \log(x)+2$
SOLUTION	$y = (C_1+C_2 \log x) * x + \log x + 2$

Example 2: Solve $x^2 y'' + x y' + 9y = \sin(3 \log(x)) + 3x^2$

S1	Cauchy's DE. Put $t=\log(x)$, $e^t=x$. $xy'=Dy$, $x^2 y''=D(D-1)y$ $D(D-1)y + Dy + 9y = \sin(3t) + 3e^{2t}$ $(D^2-D+D+9)y = \sin 3t + 3e^{2t}$ $(D^2+9)y = \sin 3t + 3e^{2t}$... (2)
CF	A.E.: $m^2+9=0 \Rightarrow m=\pm 3i$ CF = $C_1 \cos(3t) + C_2 \sin(3t)$
P1	For $\sin 3t$: $D^2 \rightarrow -9$. $f(-9) = -9+9=0$ [FAILURE!] $P_1 = t*(1/2D)*\sin(3t) = (t/2)*\int \sin(3t)dt = (t/2)*(-\cos 3t/3) = -t*\cos(3t)/6$
P2	For $3e^{2t}$: $D \rightarrow 2$. $f(2) = 4+9=13$ $P_2 = 3e^{2t}/13$

S3	y in t: $y = C1*\cos 3t + C2*\sin 3t - t*\cos 3t/6 + 3e^{(2t)}/13$ Back-sub: $t=\log x, e^t=x \Rightarrow e^{(2t)}=x^2, \cos(3t)=\cos(3\log x), \sin(3t)=\sin(3\log x)$
SOLUTION	$y = C1*\cos(3\log x) + C2*\sin(3\log x) - (\log x/6)*\cos(3\log x) + 3x^2/13$

Example 3: Solve $x^2*y'' - 3x*y' + 4y = (1+x)^2$

S1	Cauchy's: $xy'=Dy, x^2*y''=D(D-1)y, t=\log x, e^t=x$ $D(D-1)y - 3Dy + 4y = (1+e^t)^2 = 1 + 2e^t + e^{(2t)}$ $(D^2-4D+4)y = 1 + 2e^t + e^{(2t)} \dots(2)$
CF	$m^2-4m+4=0 \Rightarrow (m-2)^2=0 \Rightarrow m=2,2$ $CF = (C1+C2*t)*e^{(2t)}$
PI	PI = P1+P2+P3 $P1 = 1/(D^2-4D+4)*e^{(0t)}$. $D \rightarrow 0: f(0)=4$. $P1=1/4$ $P2 = 2e^t/(D^2-4D+4)$. $D \rightarrow 1: f(1)=1-4+4=1$. $P2=2e^t$ $P3 = e^{(2t)}/(D^2-4D+4)$. $D \rightarrow 2: f(2)=0$ FAIL. $f'(D)=2D-4$. $f'(2)=0$ FAIL again. $P3 = t^2*e^{(2t)}/2$
S3	y(t) = (C1+C2*t)*e^{(2t)} + 1/4 + 2e^t + t^2*e^{(2t)}/2 Back-sub: $t=\log x, e^t=x \Rightarrow e^{(2t)}=x^2$
SOLUTION	$y = (C1+C2*\log x)*x^2 + 1/4 + 2x + x^2*(\log x)^2/2$

4.4 Worked Examples — Legendre's Equation

Example 4: Solve $(1+x)^2*y'' + (1+x)*y' + y = 2*\sin[\log(1+x)]$

S1	Legendre's: $a=1, b=1$. Put $t=\log(1+x), e^t=1+x$ $(1+x)*y' = 1*D*y = Dy$ $(1+x)^2*y'' = 1^2*D(D-1)*y = D(D-1)y$
S2	Substitute: $D(D-1)y + Dy + y = 2\sin t$ $(D^2-D+D+1)y = 2\sin t$ $(D^2+1)y = 2\sin t \dots(2)$
CF	$m^2+1=0 \Rightarrow m=\pm i \Rightarrow CF = e^0[C1*\cos t + C2*\sin t] = C1*\cos t + C2*\sin t$
PI	$D^2 \rightarrow -1: f(-1)=-1+1=0$ [FAILURE!] $PI = t*(1/2D)*\sin t = (t/2)*(-\cos t) = -t*\cos t$
S3	y(t) = C1*cos t + C2*sin t - t*cos t Back-sub: $t=\log(1+x), \cos(t)=\cos[\log(1+x)], \sin(t)=\sin[\log(1+x)]$
SOLUTION	$y = C1*\cos[\log(x+1)] + C2*\sin[\log(x+1)] - \log(x+1)*\cos[\log(x+1)]$

Example 5: Solve $(3x+2)^2 y'' + 3(3x+2)y' - 36y = 8x^2 + 4x + 1$

S1	Legendre's: $a=3, b=2$. Put $t=\log(3x+2)$, $e^t=3x+2$, $x=(e^t-2)/3$ $(3x+2)y' = 3Dy$; $(3x+2)^2 y'' = 9D(D-1)y$
S2	Substitute: $9D(D-1)y + 9Dy - 36y = 8x^2 + 4x + 1$... RHS in terms of t : $(3x+2)^2 = e^{2t}$, so $8x^2 + 4x + 1$: $x = (e^t - 2)/3$, $3x = e^t - 2$ $(3x+2) = e^t$, $8x^2 + 4x + 1 = (8/9)(e^{2t} - 4e^t + 4) + (4/3)(e^t - 2) + 1 =$ $(8e^{2t}/9 - 32e^t/9 + 32/9) + (4e^t/3 - 8/3) + 1$ Simplifying: $= 8e^{2t}/9 + (-32/9 + 12/9)e^t +$ $(32/9 - 24/9 + 9/9) = 8e^{2t}/9 - 20e^t/9 + 17/9$ $(9D^2 - 36)y = (8/9)e^{2t} - (20/9)e^t + 17/9$ $(D^2 - 4)y = (8/81)e^{2t} - (20/81)e^t + 17/81$
CF	$m^2 - 4 = 0 \Rightarrow m = \pm 2 \Rightarrow CF = C_1 e^{2t} + C_2 e^{-2t}$
P1	e^{2t}: $D \rightarrow 2$, $f(2) = 0$ FAIL. $f'(D) = 2D$, $f'(2) = 4$ $P_1 = t \cdot (8/81) \cdot e^{2t} / (4) = (2/81) t \cdot e^{2t}$
P2	e^t: $D \rightarrow 1$, $f(1) = 1 - 4 = -3$ $P_2 = (20/81) \cdot e^t / (-3) = 20e^t / 243$
P3	$(17/81) \cdot e^0$: $D \rightarrow 0$, $f(0) = -4$ $P_3 = (17/81) / (-4) = -17/324$
SOLUTION (in t)	$y = C_1 e^{2t} + C_2 e^{-2t} + (2t/81) e^{2t} + 20e^t/243 - 17/324$

Back-substitute: $t = \log(3x+2)$, $e^t = (3x+2)$, $e^{2t} = (3x+2)^2$:

FINAL SOLUTION	$y = C_1 (3x+2)^2 + C_2 / (3x+2)^2 + [2 \log(3x+2) / 81] (3x+2)^2 + 20(3x+2) / 243 - 17/324$
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5. METHOD OF VARIATION OF PARAMETERS (VOP)

This is a general method to find the Particular Integral for any 2nd-order linear ODE $f(D)y = \phi(x)$, regardless of the form of $\phi(x)$. It works when $\phi(x)$ is NOT one of the standard types (e.g., $\sec x$, $\log x$, $\tan x$, etc.) where operator methods fail.

5.1 Working Rule — Step by Step

S1

Write the ODE in standard form and find CF

$f(D)y = \phi(x)$. Solve $f(m)=0$ to get m_1, m_2 . CF = $C_1*y_1 + C_2*y_2$ where y_1, y_2 are two independent solutions

S2

Assume the complete solution has the form

$y = A*y_1 + B*y_2$ where $A = A(x)$ and $B = B(x)$ are functions of x to be determined

S3

Compute the Wronskian W

$W = |y_1 \ y_2| = y_1*y_2' - y_2*y_1' \quad |y_1' \ y_2'|$

S4

Find A and B using the formulas

$A' = -y_2*\phi(x)/W \Rightarrow A = \int -y_2*\phi(x)/W \, dx$ $B' = y_1*\phi(x)/W \Rightarrow B = \int y_1*\phi(x)/W \, dx$

S5

Write the complete General Solution

$y = CF + PI = C_1*y_1 + C_2*y_2 + A*y_1 + B*y_2$ (The arbitrary constants from CF and the particular functions A, B from VOP combine)

5.2 Wronskian W and Formulas for A, B

WRONSKIAN

$W = y_1*y_2' - y_2*y_1'$ (must be nonzero for y_1, y_2 to be independent solutions)

FORMULA FOR A

$A = \int [-y_2 * \phi(x) / W] \, dx$

FORMULA FOR B

$B = \int [y_1 * \phi(x) / W] \, dx$

NOTE: $PI = A*y_1 + B*y_2$ where A and B are found by integration (ignoring the arbitrary integration constants — they get absorbed into CF). The method always works but may require complex integrations.

5.3 Worked Examples — Variation of Parameters

Example 1: Solve $y'' + a^2y = \sec(ax)$ by VOP

CF	A.E.: $m^2 + a^2 = 0 \Rightarrow m = \pm ia \Rightarrow CF = C_1 \cos(ax) + C_2 \sin(ax)$ $y_1 = \cos(ax), y_2 = \sin(ax)$
W	$W = y_1 y_2' - y_2 y_1' = \cos(ax)(a \cos(ax)) - \sin(ax)(-a \sin(ax))$ $= a \cos^2(ax) + a \sin^2(ax) = a[\cos^2 + \sin^2] = a$
A	$\phi(x) = \sec(ax). A' = -y_2 \phi / W = -\sin(ax) \sec(ax) / a$ $A' = -\sin(ax) / (a \cos(ax)) = -(1/a) \tan(ax) \quad A = -(1/a) \int \tan(ax) dx =$ $-(1/a) (-\log(\cos(ax)) / a) = \log(\cos(ax)) / a^2$
B	$B' = y_1 \phi / W = \cos(ax) \sec(ax) / a = 1/a$ $B = \int [1/a] dx = x/a$
PI	$PI = A y_1 + B y_2 = [\log(\cos(ax)) / a^2] \cos(ax) + [x/a] \sin(ax)$
SOLUTION	$y = C_1 \cos(ax) + C_2 \sin(ax) + (\cos(ax) / a^2) \log(\cos(ax)) + (x/a) \sin(ax)$

Example 2: Solve $y'' + 4y = \tan(2x)$ by VOP

CF	$m^2 + 4 = 0 \Rightarrow m = \pm 2i \Rightarrow CF = C_1 \cos(2x) + C_2 \sin(2x)$ $y_1 = \cos(2x), y_2 = \sin(2x)$
W	$W = 2 \cos^2(2x) + 2 \sin^2(2x) = 2$
A	$A' = -\sin(2x) \tan(2x) / 2 = -\sin^2(2x) / (2 \cos(2x))$ $= -(1 - \cos^2(2x)) / (2 \cos(2x)) = (-\sec(2x) + \cos(2x)) / 2 \quad A = (1/2) \int [\cos(2x) - \sec(2x)] dx$ $A = (1/2) [\sin(2x)/2 - \log \sec(2x) + \tan(2x) / 2] \quad A = (\sin 2x - \log \sec 2x + \tan 2x) / 4$
B	$B' = \cos(2x) \tan(2x) / 2 = \sin(2x) / 2$ $B = \int [\sin(2x) / 2] dx = -\cos(2x) / 4$
PI	$PI = A \cos(2x) + B \sin(2x)$
SOLUTION	$y = C_1 \cos 2x + C_2 \sin 2x + \cos(2x) [\sin 2x - \log \sec 2x + \tan 2x] / 4 - \sin 2x \cos 2x / 4$

Example 3: Solve $y'' - 6y' + 9y = e^{(3x)}/x^2$ by VOP

CF	$m^2 - 6m + 9 = 0 \Rightarrow (m-3)^2 = 0 \Rightarrow m = 3, 3$ $CF = (C_1 + C_2 x) e^{(3x)}$. So $y_1 = e^{(3x)}, y_2 = x e^{(3x)}$
Derivatives	$y_1' = 3e^{(3x)}, y_2' = e^{(3x)} + 3x e^{(3x)} = (1 + 3x) e^{(3x)}$

W	$W = e^{(3x)}(1+3x)e^{(3x)} - x^3e^{(3x)}$ $= e^{(6x)}[(1+3x)-3x] = e^{(6x)}$
A	$\phi(x)=e^{(3x)}/x^2$. $A'=-y^2\phi/W = -x^3e^{(3x)}e^{(3x)}/(x^2e^{(6x)})$ $= -x^3e^{(6x)}/(x^2e^{(6x)}) = -1/x$ $A = -\log(x)$
B	$B'=y^1\phi/W = e^{(3x)}e^{(3x)}/(x^2e^{(6x)}) = 1/x^2$ $B = \text{integral}[x^{(-2)}]dx = -1/x$
PI	$PI = (-\log x)e^{(3x)} + (-1/x)x^3e^{(3x)} = -e^{(3x)}\log x - e^{(3x)}$ The $-e^{(3x)}$ term is absorbed into CF. $PI = -e^{(3x)}\log(x)$
SOLUTION	$y = (C_1+C_2x) * e^{(3x)} - e^{(3x)} * \log(x)$

Example 4: Solve $y'' + y = \sec(x)\tan(x)$ by VOP

CF	$m^2+1=0 \Rightarrow m=\pm i \Rightarrow y_1=\cos x, y_2=\sin x, W=1$
A	$A'=-\sin x \sec x \tan x / 1 = -\sin x \sin x / \cos x = -\sin^2 x / \cos x$ $= -(1-\cos^2 x) / \cos x = \cos x - \sec x$ $A = \sin x - \log \sec x + \tan x $
B	$B'=\cos x \sec x \tan x = \tan x \Rightarrow B = \log \sec x + \text{const}$ Actually $B = -\log \cos x $
SOLUTION	$y = C_1 \cos x + C_2 \sin x + \sin x \cos x + \sin x - x \cos x - \sin x \log \cos x $

Example 5: Solve $y'' + 2y' + 2y = e^{(-x)}\sec^3(x)$ by VOP

CF	$m^2+2m+2=0 \Rightarrow m=(-2\pm\sqrt{4-8})/2=-1\pm i$ $CF=e^{(-x)}[C_1\cos x+C_2\sin x]$. $y_1=e^{(-x)}\cos x, y_2=e^{(-x)}\sin x$
W	$W = e^{(-2x)}[\cos x(\cos x - \sin x) - \sin x(-\sin x - \cos x)]$ $= e^{(-2x)}[\cos^2 x - \sin x \cos x + \sin^2 x + \sin x \cos x] = e^{(-2x)}$
A	$A'=-e^{(-x)}\sin x * e^{(-x)}\sec^3 x / e^{(-2x)} = -\sin x / \cos^3 x = -\tan x \sec^2 x$ $A = \text{integral}[-\tan x \sec^2 x]dx$. Let $t=\tan x, dt=\sec^2 x dx$ $A = -\text{integral}[t*dt] = -\tan^2 x / 2$
B	$B'=e^{(-x)}\cos x * e^{(-x)}\sec^3 x / e^{(-2x)} = \cos x \sec^3 x = \sec^2 x$ $B = \text{integral}[\sec^2 x]dx = \tan x$
PI	$PI = (-\tan^2 x / 2) * e^{(-x)} * \cos x + \tan x * e^{(-x)} * \sin x$ $= e^{(-x)} * \cos x * (-\tan^2 x / 2) + e^{(-x)} * \sin x * \tan x = e^{(-x)} * \cos x * (-\tan^2 x / 2 + \tan x \sin x / \cos x) =$ $e^{(-x)} * [-\sin^2 x / (2 \cos x) + \sin^2 x / \cos x] = e^{(-x)} * \sin^2 x / (2 \cos x) = (e^{(-x)} / 2) * \sin x * \tan x$
SOLUTION	$y = e^{(-x)} [C_1 \cos x + C_2 \sin x] + (e^{(-x)} / 2) * \sin x * \tan x$

6. APPLICATIONS — L-C AND L-C-R CIRCUITS

Linear ODEs with constant coefficients model electrical circuits. The charge $q(t)$ on a capacitor or the current $i(t)$ through an inductor satisfies a linear ODE whose order equals the number of energy-storing elements.

6.1 L-C Circuit (Pure Inductance-Capacitance — Undamped)

In a simple L-C circuit with no resistance ($R=0$), the charge $q(t)$ satisfies:

L-C CIRCUIT ODE

$$L \frac{d^2 q}{dt^2} + \frac{q}{C} = E(t) \quad \text{where } i = \frac{dq}{dt}$$

For $E(t) = E \sin(\omega t)$ and natural frequency $\omega_0^2 = 1/(LC)$:

AE

Characteristic equation: $Lm^2 + 1/C = 0$

$m^2 = -1/(LC) = -\omega_0^2$ $m = \pm i\omega_0$ (purely imaginary roots) CF = $C_1 \cos(\omega_0 t) + C_2 \sin(\omega_0 t)$

PI

For $E \sin(\omega t)$ with $\omega \neq \omega_0$: $D^2 \rightarrow -\omega^2$

PI = $E \sin(\omega t) / (L(-\omega^2) + 1/C) = E \sin(\omega t) / (1/C - L\omega^2) = EC \sin(\omega t) / (1 - LC\omega^2)$

NOTE: RESONANCE: If $\omega = \omega_0$ (forcing frequency = natural frequency), the denominator = 0 and PI involves the failure case. This gives oscillations that grow without bound — physically this is electrical resonance.

6.2 L-C-R Circuit (Series Circuit — Damped Oscillations)

In a series L-C-R circuit with EMF $E \sin(\omega t)$, the charge $q(t)$ satisfies:

L-C-R ODE

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = E \sin(\omega t)$$

This is a 2nd-order non-homogeneous ODE with constant coefficients.

AE

$m^2 + (R/L)m + 1/(LC) = 0$

$m = -R/(2L) \pm \sqrt{(R/(2L))^2 - 1/(LC)}$ Let $p^2 = 1/(LC) - R^2/(4L^2)$ [for underdamped case]
 $m = -R/(2L) \pm i p$

CF

CF = $e^{-(Rt/2L)} * [C_1 \cos(pt) + C_2 \sin(pt)]$

This represents the transient (decaying) part of the response

PI

$D^2 \rightarrow -\omega^2$ in $(LD^2 + RD + 1/C)$:

$L(-\omega^2) + R^2 D + 1/C$ is NOT purely numerical (still has D term) PI = $E \sin(\omega t) / (LD^2 + RD + 1/C)$ $\{D^2 \rightarrow -\omega^2\}$ At resonance $\omega^2 = 1/(LC)$: denominator reduces to RD PI = $E \sin(\omega t) / (RD) = (E/R) \int \sin(\omega t) dt = -E \cos(\omega t) / (R\omega)$

6.3 Resonance Condition — LCR Circuit Derivation

VTU STANDARD PROBLEM: In an L-C-R circuit, with $\omega^2 = 1/(LC)$, initially $q=0$ and $i=0$ when $t=0$. Show that:

**RESONANCE
FORMULA**

$$q = (E/R*\omega) * e^{(-Rt/2L)} [\cos(pt) + (R/2pL) * \sin(pt)] - (E/R*\omega) * \cos(\omega t)$$

S1	ODE: $(LD^2 + RD + 1/C)q = E \sin(\omega t)$. Divide by L: $(D^2 + (R/L)D + 1/(LC))q = (E/L) \sin(\omega t)$
S2	A.E.: $m^2 + (R/L)m + 1/(LC) = 0$. Using quadratic formula and $p^2 = 1/(LC) - R^2/(4L^2)$: $m = -R/(2L) \pm ip$ CF = $e^{(-Rt/2L)}[C_1 \cos(pt) + C_2 \sin(pt)]$
S3	PI: Since $\omega^2 = 1/(LC)$ [resonance], $(L*(-\omega^2) + 1/C) = 0$, only RD remains: $PI = (E/L) \sin(\omega t) / (RD/L + 0) = E \sin(\omega t) / (RD) = (E/R) * (1/D) \sin(\omega t) = (E/R) * (-\cos(\omega t) / \omega)$ PI = $-E \cos(\omega t) / (R \omega)$
S4	Apply IC: $q(0) = 0 \Rightarrow C_1 = E/(R \omega)$ Apply IC: $i = dq/dt = 0$ at $t = 0 \Rightarrow C_2 = E R / (2pL * R \omega) = E / (2pL \omega)$
S5	Substitute C1, C2 and simplify by taking $E/(R \omega)$ common:
VERIFIED	$q = (E/R*\omega) * \{ e^{(-Rt/2L)} [\cos(pt) + (R/2pL) * \sin(pt)] - \cos(\omega t) \}$

7. PREVIOUS YEAR VTU QUESTIONS WITH DETAILED SOLUTIONS

NOTE: These are actual VTU exam questions. Mastering all of them guarantees exam readiness.

7.1 Dec 2019/Jan 2020 — Q1 (10 Marks)

Solve: $(D^2-4D+13)y = e^{(3x)} \cosh(2x) + 3^x$

CF	$m^2-4m+13=0$. $m=(4\pm\sqrt{16-52})/2=(4\pm6i)/2=2\pm3i$ CF = $e^{(2x)}[C1*\cos(3x)+C2*\sin(3x)]$
PI-P 1	$\cosh(2x)=(e^{(2x)}+e^{(-2x)})/2 \Rightarrow e^{(3x)} \cosh 2x=(e^{(5x)}+e^x)/2$ $P1=(1/2)*[e^{(5x)}/(25-20+13) + e^x/(1-4+13)] = (1/2)[e^{(5x)}/18+e^x/10]$
PI-P 2	$3^x=e^{(x*\log 3)}$. $D \rightarrow \log 3$ $P2=3^x/[(\log 3)^2-4(\log 3)+13]$
ANS WER	$y=e^{(2x)} [C1*\cos 3x+C2*\sin 3x] + e^{(5x)}/36 + e^x/20 + 3^x/[(\log 3)^2-4\log 3+13]$

7.2 June/July 2019 — Q2 (10 Marks)

Solve: $(D^2+3D+2)y = 4\cos^2(x)$

Rewrite: $4\cos^2(x) = 2(1+\cos 2x) = 2e^{(0x)} + 2\cos(2x)$

CF	$m^2+3m+2=0 \Rightarrow (m+1)(m+2)=0 \Rightarrow m=-1, -2$ CF = $C1*e^{(-x)} + C2*e^{(-2x)}$
P1	$2*e^{(0x)}/(D^2+3D+2)$. $D \rightarrow 0$: $f(0)=2$. $P1=2/2=1$
P2	$2\cos(2x)/(D^2+3D+2)$. $D^2 \rightarrow -4$: $-4+3D+2=3D-2$ Rationalize: $(3D+2)/(9D^2-4)$. $D^2 \rightarrow -4$: $9(-4)-4=-40$ $P2=[3D(2\cos 2x)+2*2\cos 2x]/(-40) = [-12\sin 2x+4\cos 2x]/(-40) = (3\sin 2x-\cos 2x)/10$
ANS WER	$y = C1*e^{(-x)} + C2*e^{(-2x)} + 1 + (3\sin 2x - \cos 2x)/10$

7.3 Dec 2018/Jan 2019 — Q3 (10 Marks)

Solve: $x^2 y'' - x y' + y = \log(x)$ (Cauchy's ODE)

Sub	$t=\log x$, $e^t=x$. $xy'=Dy$, $x^2 y''=D(D-1)y$ $\Rightarrow D(D-1)y-Dy+y=t \Rightarrow (D^2-2D+1)y=t$
CF	$(m-1)^2=0 \Rightarrow m=1, 1$. CF = $(C1+C2*t)*e^t$

PI	$PI = t/(1-2D+D^2) = t/(1+(-2D+D^2)) = (1+2D+3D^2+...)*t = t+2$
ANS WER	$y = (C_1 + C_2 \log x) * x + \log x + 2$

7.4 May/June 2018 — Q4 (10 Marks)

Solve by VOP: $y'' + y = \sec(x) \tan(x)$

CF	$m^2 + 1 = 0 \Rightarrow m = \pm i$. $y_1 = \cos x$, $y_2 = \sin x$, $W = 1$
A	$A' = -\sin x \sec x \tan x = -(\sin x / \cos x) * (\sin x / \cos x) * \cos x = -\sin^2 x / \cos x$ $= -(1 - \cos^2 x) / \cos x = \cos x - \sec x$ $A = \sin x - \log \sec x + \tan x $
B	$B' = \cos x \sec x \tan x = \tan x \Rightarrow B = -\log \cos x = \log \sec x $
ANS WER	$y = C_1 \cos x + C_2 \sin x + (\sin x - \log \sec x + \tan x) * \cos x + \sin x * \log \sec x $

7.5 Dec 2017/Jan 2018 — Q5 (10 Marks)

Solve: $(1+x)^2 y'' + (1+x)y' + y = 2 \sin[\log(1+x)]$ (Legendre's ODE)

Sub	$t = \log(1+x)$. $(1+x)y' = Dy$. $(1+x)^2 y'' = D(D-1)y$ $\Rightarrow D(D-1)y + Dy + y = 2 \sin t \Rightarrow (D^2 + 1)y = 2 \sin t$
CF	$m^2 + 1 = 0 \Rightarrow m = \pm i$. $CF = C_1 \cos t + C_2 \sin t$
PI	$D^2 \rightarrow -1$: $f(-1) = 0$ FAIL. $PI = -t \cos t / 2 = -t \cos t$
ANS WER	$y = C_1 \cos[\log(1+x)] + C_2 \sin[\log(1+x)] - \log(1+x) \cos[\log(1+x)]$

7.6 June/July 2016 — Q6 (10 Marks)

Solve $y'' - 6y' + 9y = e^{(3x)}/x^2$ by Variation of Parameters

CF	$(m-3)^2 = 0 \Rightarrow m = 3, 3$. $CF = (C_1 + C_2 x) * e^{(3x)}$. $y_1 = e^{(3x)}$, $y_2 = x * e^{(3x)}$
W	$W = e^{(6x)}$ (computed above)
A	$A = -\log x$ (computed above). $B = -1/x$ (computed above)

ANS WER	$y = (C_1 + C_2 x) e^{(3x)} - e^{(3x)} \log x$

7.7 Dec 2016/Jan 2017 — Q7 (10 Marks)

Solve: $(D^4 - m^4)y = 0$ and find CF

Roots of $p^4 = m^4$: $p = m, -m, +mi, -mi$

ANSWER	$y = C_1 e^{(mx)} + C_2 e^{(-mx)} + C_3 \cos(mx) + C_4 \sin(mx)$
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7.8 Dec 2020 — Q8 (10 Marks)

Solve $(D^3 - 1)y = 3\cos(2x) + e^{(-2x)}$

CF	$m^3 - 1 = 0 \Rightarrow (m-1)(m^2 + m + 1) = 0 \Rightarrow m = 1, m = \frac{-1 \pm i\sqrt{3}}{2}$ CF = $C_1 e^x + e^{(-x/2)} [C_2 \cos(\sqrt{3}x/2) + C_3 \sin(\sqrt{3}x/2)]$
P1	$3\cos(2x)/(D^3 - 1)$. $D^3 \rightarrow D^* D^2 \rightarrow D^*(-4) = -4D$ at numerator level. $f(D) = D^3 - 1$. For $\cos 2x$: replace $D^2 \rightarrow -4$ where possible: $D^3 = D^* D^2 = D^*(-4) = -4D \Rightarrow$ $f(D) \cos = -4D - 1$ P1 = $3\cos(2x)/(-4D - 1)$. Multiply by $(-4D + 1)/(-4D + 1)$: $= \frac{3(-4D + 1)\cos(2x)}{(16D^2 - 1)}$. $D^2 \rightarrow -4$: $16(-4) - 1 = -65$ P1 = $\frac{3(-4D(\cos 2x) + \cos 2x)}{(-65)} =$ $\frac{3(8\sin 2x + \cos 2x)}{(-65)}$ P1 = $-\frac{3(8\sin 2x + \cos 2x)}{65}$
P2	$e^{(-2x)}/(D^3 - 1)$. $D \rightarrow -2$: $f(-2) = -8 - 1 = -9$ P2 = $e^{(-2x)}/(-9) = -e^{(-2x)}/9$
ANS WER	$y = C_1 e^x + e^{(-x/2)} [C_2 \cos(\sqrt{3}x/2) + C_3 \sin(\sqrt{3}x/2)] - \frac{3(8\sin 2x + \cos 2x)}{65} - e^{(-2x)}/9$

8. QUICK REVISION SUMMARY

Use this section for last-minute exam preparation. All key procedures at a glance.

8.1 Complete Solution Strategy for any Linear ODE

Step	Action	Detail
1	Check ODE type	Constant coeff? $\Rightarrow$ operator method. Variable coeff (x^n)? $\Rightarrow$ Cauchy's. $(ax+b)^n$? $\Rightarrow$ Legendre's. VOP? $\Rightarrow$ when operator fails
2	Find CF	Write A.E. $f(m)=0$. Solve roots. Apply Case I/II/III rules.
3	Find PI	Identify $\phi(x)$ type: e^{ax} , $\sin/\cos$, polynomial, or $e^{ax} \cdot V$. Apply operator rule. Check for failure cases ($\text{denom}=0$).
4	Write GS	$y = \text{CF} + \text{PI}$. For IVP: apply initial conditions to find $C_1, C_2, \dots$
5	Cauchy's	Put $t=\log x$; use $xy'=Dy$, $x^2y''=D(D-1)y$. Solve const-coeff ODE in t . Back-sub.
6	Legendre's	Put $t=\log(ax+b)$; use $(ax+b)y'=aDy$, $(ax+b)^2y''=a^2D(D-1)y$. Same as Cauchy's.
7	VOP	Find y_1, y_2 from CF. Compute W . Find $A=-\int(y_2\phi/W)dx$, $B=\int(y_1\phi/W)dx$. $\text{PI}=Ay_1+By_2$.

8.2 PI Failure Case Quick Reference

NOTE: FAILURE happens when the substitution $D=a$ gives $f(a)=0$ (denominator vanishes). Each failure introduces one additional factor of x and differentiates the denominator.

Failure Condition	Formula to Use
$f(a)=0$ [first failure]	$\text{PI} = x \cdot e^{ax} / f'(a)$ [differentiate denom]
$f'(a)=0$ [second failure]	$\text{PI} = x^2 \cdot e^{ax} / f''(a)$
$f(-a^2)=0$ for $\sin/\cos$	$\text{PI} = x \cdot \sin(ax) / 2f'(-a^2)$ [similar rule]
$f(0)=0$ for polynomial type	$\text{PI} = x \cdot \phi(x) / f'(D) _{D=0}$

8.3 Exam Strategy Tips

- Always write the A.E. explicitly before solving it — examiners check this step
- For Cauchy's / Legendre's: write the transformation formulas before substituting
- After finding PI, verify by substituting back into the ODE (if time permits)
- For VOP: always compute W FIRST — if $W=0$, your y_1, y_2 are NOT independent (error in CF)
- Complex roots $\alpha \pm i\beta$: write CF as $e^{\alpha x} [C_1 \cos(\beta x) + C_2 \sin(\beta x)]$ — do NOT write $C_1 e^{(\alpha + i\beta)x}$
- Failure case: each 'failure' multiplies PI by x . Two failures $\Rightarrow x^2$. Three $\Rightarrow x^3$
- For 3^x : convert $3^x = e^{(x \log 3)}$ before applying operator method
- For $\cosh/\sinh$: ALWAYS convert to exponentials using definitions
- For $4\cos^2(x)$: ALWAYS convert using $\cos^2 = (1 + \cos 2x)/2$ first

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